
Some Aspects of Charge Transfer in Biological Systems

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ABSTRACT: The following modes of biological charge transfer are discussed: via solitons, as electronic conductance as hydrated proteins, via surface charge transfer complexes, and by protonic charge transfer complexes. Electron transfer involving the cytochromes is treated in the light of electrochemical measurements including voltammetry. A hypothesis for the lateral, surface conductance of biological membranes is proposed, based on electron donor-acceptor surface interactions giving rise to polaritons.

1. Introduction

In a way, all biological charge transfer is electrochemical in nature. Thus, this field is so huge that the following discussion will have to be confined to a few selected aspects. Many relevant topics are treated by the other contributors to this volume.

Regular and periodic organization of molecules is essential for the living state, and fundamental to its processes are structured instabilities which are reflected in the solid state by, e.g., charge transfer complexes. Many *in vivo* properties are difficult to explain by classical chemical processes in solution but appear to fit solid state physical processes in cells. Solid state events involving conduction are evident in animate aggregations and may well be an essential characteristic of life, which may be an electromagnetic phenomenon. A growing body of reviews and texts⁽¹⁾ is

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available to support these views. Chelate metal complexes of purines, pyrimidines, and their nucleosides have been reviewed in detail.⁽²⁾

For charge transfer to occur, free charges must either be initially, inherently present, or injected from outside, or else be generated within the system by means of an injection of energy.

When a molecule is excited by a sufficiently energetic event, an electron can be ejected. The resulting molecular cation polarizes electrons from its immediate environment in a time of about 10^{-16} s, and this polarization energy stabilizes the final state of the system.

When two opposite charges in a medium of relative permittivity ϵ_r approach one another, the Coulombic energy of attraction between them equals kT at a critical distance R_c so that⁽³⁾

$$R_c = e^2/4\epsilon_r\epsilon_0kT$$

or

$$R_c = 55.9/e_r \text{ nm}$$

(1)

A limit therefore exists for the quantity of charge of one kind which can be imposed without resulting in recombination.⁽³⁾ This limits the amounts to about 10^{17} electronic charges/cm³, while there are about 10^{21} molecules/cm³. Electric neutrality need not be preserved in organic media. In an organic layer of about 5 nm the total of such uncompensated space charge is approximately 10^{-10} C/cm², which, if it were uniformly distributed, would yield a rather small electric field of about 10^3 V cm⁻¹. Potential differences of about 5 mV across 5 nm, i.e., 10^4 V cm⁻¹, do not represent a serious problem for space charge limitation of current flow through biological systems typically several nanometers thick. It also follows that only one molecule in several thousand can be charged if compensating charges are absent. Thus, modeling a biological system, a localized representation based on individual charged species is preferable to a representation of a charge density distributed over all the molecules.⁽³⁾

If the distance between M^+ and M^- is less than R_c , they exist as an ion pair, i.e., in a charge transfer state. If M^+ and M^- are locked into original lattice states then the distance can have only certain values. From the energies for linear polyacenes, determined accurately by Silinsh,⁽⁴⁾ it appears that, as the number of rings increased,* charge separation in the

* Silinsh estimated⁽⁴⁾ the energy (E_{CT}) of the charge transfer state of pentacene to be about one order of magnitude smaller than that of naphthalene. As the number of conjugated rings increases it becomes easier to separate the charges in a CT state; the photoinduced M^+M^- thermalized CT state of pentacene has, under some conditions, charge separation distances as large as 12.5 nm.⁽⁵⁾

charge transfer state is facilitated. This could be significant for biological systems containing highly conjugated molecules such as chlorophyll.⁽³⁾

Electrochemical methods have been applied to the study of a variety of biological charge transfer problems. Thus, to give just a few examples, voltammetry has been used to study tissue,⁽⁶⁾ zeta potentials have been related to cystic fibrosis,⁽⁷⁾ an electrophoretic mobility test of macrophage for malignancy has been suggested⁽⁸⁾ and polarography has been applied, *inter alia*, to the study of carcinogenicity of several chemical compounds by the Czech school.⁽⁹⁾

2. Solitons⁽⁷³⁾

The important role played by solitons* in biological transfer remains to be clarified: solitons are said⁽¹¹⁾ to arise from localized disordered regions on, say, a membrane surface in the presence of the transmembrane electric field; the local perturbation then tends to spread and to move leading to changes in the orientation of the lipid membrane molecules. The soliton energy is considerably below the "energy band gap" of a polypeptide chain but may initiate proton transfer⁽¹²⁾ in a hydrogen bonded chain in the presence of an electrostatic field.

Solitons may result in energy transporting along biological macromolecules. Their genesis and propagation as phonons in macromolecules establish important system properties in these molecules. Electromagnetic energy may be converted to soliton conducted energy as a transductive step. These solitons propagate quite slowly along the molecule, exhibiting threshold and possibly "windowed" characteristics in relation to the exciting energy. In transductive coupling of exciting events on cell membranes, they thus offer a vehicle for highly selective relationships in the electrochemical environment. In the transmembrane coupling of cell surface events to specific intracellular enzyme systems or organelles, either at internal membrane sites or via the cytoskeletal system to submembrane regions, solitons may also play a major role.

Binding of drug molecules to helical protein molecules⁽⁶⁷⁾ may inhibit soliton propagation. Further studies along these lines may allow "tailoring" of new drugs with optimal therapeutic capabilities. Their actions may be modulated by concurrent exposure to selected electromagnetic fields. Similar selective modulations of tissue functions by modification of the

* A soliton⁽¹⁰⁾ is a localized collective excited state likely to arise in nonlinear dispersive media. The soliton is capable of migrating over relatively long distances with very little energy dissipation. It may be electrically charged or may be neutral such as a kink in a polymeric chain, or a domain wall.

electromagnetic environment may be seen in wound healing, tissue regeneration, and cancer therapy. These studies in biological interactions with the electromagnetic environment may yield important new knowledge about biophysical bases of circadian and other biological rhythms, with important therapeutic implications in endocrine and immune disorders.

The solitons are thought⁽⁶⁸⁾ to induce simultaneously a displacement of the vibrational energy levels of the macromolecular chain and a rearrangement of the aqueous environment, until some sort of resonance is attained between these two processes, and a transition between charge and discharge regimes occurs. The result is a region of highly structured water and what often is called a Fröhlich regime, characterized by the existence of coherent electrical oscillations having frequencies of the order of 10–100 GHz, and discussed by Fröhlich himself in Chapter 8 of this volume. In other words, the metabolic energy reaches definite, localized sites within the macromolecular chain, e.g., DNA or protein. The energy causes a local deformation which travels along the chain as a soliton, causing the two effects mentioned above. Eventually, the soliton energy is transferred into the aqueous phase as a Fröhlich polarization wave. Maintained by the continuing supply of metabolic energy, this wave then plays a major part in affecting the cell organization.

One main problem still remains, viz., the isolation and characterization of the solitons as appearing in biological systems.

3. Conduction and Biological Structure

Much of what follows is based on discussions by Pethig and co-workers.⁽¹³⁾ (See his contribution to this volume, Chapter 7.)

Proteins have the general structure of polypeptide chains with $-C-C-N-$ as a repeat unit; see Figure 1.

Each peptide unit lies in a plane because it consists of a delocalized system of π electrons associated with π orbitals of the C and O atoms together with the lone electron-pair orbital of the N atom. Such an electron resonance structure is sufficient to produce significant diamagnetic anisotropy in the protein.⁽¹⁴⁾ Some two dozen amino-acid residues make up polypeptides, but only glycine and proline have a first atom in the side chain R which is not a carbon atom. Thus, many features of regularity are present which may cooperate in forming energy bands, particularly in the extended arrays of α helices and β -pleated sheets. In fact, spectra of DNA

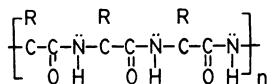


FIGURE 1. General structure of proteins.

and RNA are not basically different from those of their constituent groups because randomization of the constituents washes out all features of the electronic spectrum. Tong⁽¹⁵⁾ suggests that this is due to the presence of geometry-independent lines and allowed energy bands. Altieri and Krizan,⁽¹⁶⁾ using a self-consistent method, show term energies of reasonable magnitude to exist if DNA and related models are characterized as intrinsic semiconductors with a band energy gap of approximately 2 eV. Many proteins have a semiconduction activation energy of about this value.⁽¹⁷⁾

Whether charge carriers can migrate freely through proteins depends greatly on these energy bands. Coherent electronic motion can occur if broad bands of electronic states are available for the ground states (valency bands) and the excited states (conduction bands). If the bands of the extended states are less than about 2 kT, or if the energy states are localized, charge transport takes place by activated hopping or by tunneling processes.⁽¹³⁾

It appears that all proteins in the natural "pure" state are insulators. The valence band of the extended states is completely occupied by electrons, and the band gap is so large that at physiological temperatures the possibility of promoting electrons across the gap is negligible. Then, calculations⁽¹³⁾ indicate that the energy band widths are greater for atomic interactions along the polypeptide backbone chains ($-C-C-N-C-C-N-$) than those arising from the hydrogen-bonded network interactions which stabilize the protein's tertiary structure. Thus, water molecules in a gramicidine A channel form a linear structure along the channel axis and containing about seven molecules.⁽⁹⁶⁾ The majority of semiconduction experiments have involved soluble proteins, probably because these are easily purified and crystallized. However, the structural proteins, those which support the main functional elements, are usually discarded in purification processes, as residues. Experimental evidence has existed for some time for electronic conduction in sea-animal integuments, e.g., crab shells.⁽¹²⁾ In order to function, such structural proteins will generally be complexed with other molecules in a manner analogous to dopants and thereby facilitate charge transport.

4. Hydration and Charge Transfer

4.1. Proteins

In their natural state many proteins are bound into hydrophobic lipid matrices. The energetics of charge separation and mobility are governed largely by the dielectric constant of the surrounding medium. Calculations⁽¹³⁾ have shown that the internal structures have an effective

high-frequency relative permittivity of about 2.6, which value does not increase much upon hydration. Even in the presence of a fully extended hydrogen-bonded network, conduction along the backbone is more likely, but the particular properties of a protein will depend on its precise conformation, environment, and interactions with other molecules. Thus, it has been shown⁽¹⁹⁾ that the measured conductances and permittivities of bovine serum albumin and lysozyme are determined by the degree of hydration as well as of the NaCl concentration in the salt solution used as the medium. The electrical behavior is governed by protein–water interactions; the conductance is due to a proton site-to-site hopping process.⁽¹⁹⁾

Considerable differences in the resistivity between that of ordinary proteins and that of electron transfer proteins have been reported^(20,21) and also between ferrocycytochrome-*c* and ferricycycytochrome-*c*. This is attributed to the electronic state of the central metal atom.⁽²⁰⁾ The anhydrous cytochromes exhibit resistivities many orders of magnitude higher; thus anhydrous ferricycycytochrome *c*₃ has a room temperature of $4.1 \times 10^{12} \Omega \text{ cm}$ and the ferrocycytochrome *c*₃ a value of 1.6×10^{10} ; these data are even more remarkable because the ferri compound is said⁽²⁴⁾ not to follow an Arrhenius type relationship in its temperature dependence while the ferro compound is reported to yield a negative value for its thermal activation energy. Cytochromes will be discussed in more detail in the next section.

Another interesting example for severe hydration effects in relation to charge transfer is the purple membrane of *Halobacterium halobium*: it acts as a photoelectret, producing at least semipermanent electric charges upon illumination; this activity has been shown to be greatly affected by hydration.⁽⁹⁵⁾

Water in proteins has been determined from hydration isotherms obtained gravimetrically and the more recent quartz crystal resonator techniques.⁽³⁹⁾ The steady state conductivity of protein samples increases rapidly with water absorbed. The weight percentage, *m*, relates to the conductivity according to the Spivey equation,⁽⁴⁰⁾

$$\lambda = \lambda_D \exp(\alpha m) \quad (2)$$

where λ_D is the dry-state conductivity and α a constant. For *m* of about 5 wt %, a change occurs such that $\alpha = 1.3$ for $m < 5 \text{ wt } \%$ and $\alpha = 0.9$ for $m > 5 \text{ wt } \%$. This may be explained by assuming that at 5 wt % nearly all the protein primary sorption sites are occupied by water molecules and the population of secondary hydration sites begins. The frequency f_m of maximum loss of dielectric dispersion is also found⁽¹³⁾ to increase with increasing hydration following the relationship

$$\tau = \tau_0 \exp(-\beta m) \quad (3)$$

where τ is the relaxation time and β a constant. Again at 5 wt % the value of the constant changes.⁽¹³⁾ Although f_m changes, the area under the dielectric loss factors ε'' vs. $\log(f)$ curve does not change with hydration, suggesting⁽¹³⁾ that the observed dielectric dispersions are not directly related to the relaxation of water dipoles.

4.2. Cytochromes

Cytochrome- c_3 is specially interesting because this globular hemoprotein (molecular weight about 12,500) is a defined chemical entity and known⁽²²⁾ to be the electron carrier in the respiratory chain of, *inter alia*, *Desulfovibrio vulgaris*. The electron transfer rate in its reversible electrochemical electrode⁽⁹⁸⁾ redox reaction has the enormous value of 0.1 cm sec^{-1} . Its resistivity in the ferro- as well as the ferri-form is shown in Figure 2. The conductivity was ohmic and appears to be electronic in nature⁽²⁰⁾; values are said⁽²⁰⁾ to be well reproducible. The low-temperature branch of ferrocytochrome- c exhibits the usual Arrhenius form, but the high-temperature region shows a most unusual behavior, as seen from Figure 2. The resistivities of these two forms are reported⁽²⁰⁾ to differ by 11 orders of magnitude, with the reduced form having the extraordinarily low value of $57 \Omega \text{ cm}$ at the transition temperature T_M , a value comparable to that of germanium. It is remarkable that even anhydrous films of tetrahemoprotein cytochrome C_3 exhibit⁽²³⁾ good conductivity at 292 K.

The polypeptide chain within the cytochrome molecule has virtually no conductivity at all, while the complete hemoprotein exhibits a conductivity of the order of 10^{-9} – $10^{-11} (\Omega \text{ cm})^{-1}$, suggesting a considerable contribution of the heme units to the overall electron transfer. Since the porphyrin subunit is about $13 \text{ \AA}$ diameter, sufficient delocalization of π electrons should be expected⁽²⁰⁾ to impart conductivity. It has been

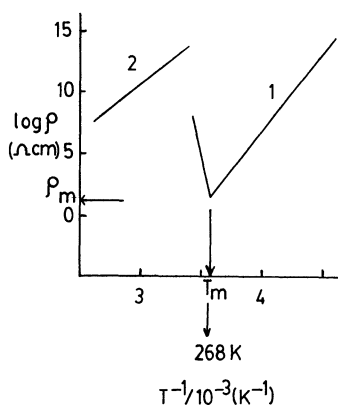


FIGURE 2. Temperature dependence of the resistivity of (1) ferrocytochrome- c_3 and (2) ferricytochrome- c_3 .⁽²⁰⁾

hypothesized that some part of the hemes in cytochrome- c_3 are exposed to the protein surface so that electrons may tunnel through a peptide residue.⁽¹⁹⁾ The heme-heme interaction in the cytochrome c_3 molecule was inferred from NMR and Mössbauer spectroscopy. Cytochrome c_3 undergoes a reversible oxidoreduction by catalytic action of hydrogenase.⁽²⁶⁾

Table 1 collects the electrical conductivity data of several protein solids.⁽²⁷⁾ The ferri-forms of both cytochromes have a similar resistivity in magnitude at room temperature. On the other hand, very drastic differences appeared in the resistivity of the ferro-form.

The contribution of heme to the electrical conduction of biological materials is clear because the polypeptide chain itself has a resistivity $10^{14} \Omega \text{ cm}$. On the other hand, the resistivity of one hemoprotein is about 10^9 – $10^{11} \Omega \text{ cm}$.

Taking into account the size of porphyrin, 1.3 nm in diameter, the distance is small enough to cause the overlapping of π electrons. Electrochemical measurement of the reduction of cytochrome c_3 on a mercury electrode also suggests⁽³¹⁾ that some parts of hemes are exposed to the protein surface and/or the electrons tunnel through a peptide residue.

Rapid direct electron transfer between cytochrome c and a gold electrode takes place in the presence of 4,4'-bipyridyl and 1,2-bis(4-pyridyl)ethylene, which form an adsorbed layer on the electrode surface thus facilitating the electron-transfer reaction.⁽³²⁾ There are some striking

TABLE 1. Electrical Conductivity of Several Protein Solids^a

Substance	Temperature range (°C)	E in $E/2kT$ (eV)	Resistivity ($\Omega \text{ cm}$) at temperature indicated (°C)
Cytochrome c oxidized ^b	—	2.7	6.1×10^{16} (30)
Cytochrome c oxidized ^b	50 to 85	1.2	3.1×10^{11} (30)
Cytochrome c reduced ^b	10 to 60	1.2	3.1×10^9 (30)
Cytochrome c_3 oxidized ^b	20 to 70	3.3	2.3×10^{12} (30)
Cytochrome c_3 reduced ^b	–40 to –5	7.7(?)	5.7×10^1 (–5)
Ferricytochrome c	—	1.2	6.5×10^{10} (55)
Ferrocycytochrome c	—	1.2	6.5×10^8 (55)
Lysozyme	—	—	$> 10^{14}$ (30)
Myoglobin	—	0.31	3.6×10^{10} (30)
Ribonuclease	—	—	$> 5 \times 10^{14}$ (30)
Trypsin	—	—	$> 10^{14}$ (30)

^a After Nakahara, Kimura, and Inokuchi.⁽³⁰⁾

^b All samples in the condensed form measured in surface cells with gold electrodes.

ing analogies between the reaction of cytochrome *c* at this electrode and its reaction with cytochrome oxidase with respect to the effect of chemical modification of the cytochrome *c* lysine residues and the effect of poly-L-lysine on the electrode reaction. The results suggest that cytochrome *c* binds to the 4,4'-bipyridyl-modified gold electrode surface prior to electron transfer in a manner similar to its interaction with the oxidase.

The electron transfer reaction between cytochrome *c* and cytochrome oxidase proceeds via a protein complex in which the ϵ -amino groups of the cytochrome *c* lysine residues are believed⁽³³⁾ to play an important role. Chemical modification of these lysines is well known⁽³⁴⁾ to affect the cytochrome *c*-oxidase electron-transfer reaction. The importance of lysine residues in the protein-protein interaction is further illustrated by the effect⁽³⁵⁾ of poly-L-lysine, a competitive inhibitor of the cytochrome *c*-oxidase reaction which acts by binding to the oxidase.⁽³⁶⁾ Both *N*-acetimidyllysyl and *N*-guanidinyllysyl horse heart cytochrome *c* are electroactive at a gold electrode in the presence of 4,4'-bipyridyl or 1,2-bis(4-pyridyl)ethylene, giving rise to quasireversible diffusion-controlled dc and ac voltammograms indistinguishable from those (Figure 3) of the native protein,⁽³⁷⁾ with a half-wave potential, $E_{1/2r} = 0.25$ V vs. NHE. Similarly⁽³⁶⁾ both are enzymatically active⁽³⁷⁾ in the cytochrome-oxidase system. The enzymatically inactive *N*-trifluoroacetyl and *N*-maleyl derivatives are both electroinactive.⁽³²⁾

Poly-L-lysine inhibits the electrode reaction of native horse heart cytochrome *c*, as shown by dc voltammetry (Figure 3), again analogous to its inhibiting effect on the cytochrome *c*-oxidase reaction. The effect on the ac cyclic voltammetry peak current, $i_p(\text{ac})$, is more marked. The variation with poly-L-lysine concentration is consistent with adsorption of poly-L-lysine onto the electrode surface, decreasing the effective free electrode area.

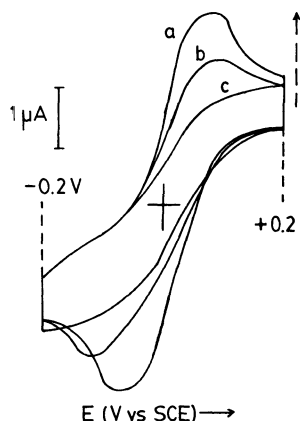


FIGURE 3. dc Voltammogram of horse heart ferri-cytochrome *c*, 0.4 nM in 0.1 M NaClO₄, 0.02 M phosphate buffer, pH 7, saturated solution of 1,2-bis(4-pyridyl)ethylene in the potential range from +0.2 to -0.2 V vs. SCE with poly-L-lysine: (a) 0 mg ml⁻¹, (b) 1 mg ml⁻¹, (c) 1.5 mg ml⁻¹, dc potential scan rate, 100 mV s⁻¹. After Eddowes *et al.*⁽³²⁾

The electron-transfer reaction of the native protein has been studied⁽³²⁾ by ac impedance measurements. The heterogeneous electron-transfer rate was found to be as fast as that determined⁽³⁸⁾ for the ferri/ferrocyanide couple ($0.3\text{--}3.0 \times 10^{-4}$ msec⁻¹).

Double-potential-step chronocoulometry, rotating disk, and ac rotating ring disk indicate⁽³²⁾ that the native cytochrome binds to the 4,4'-bipyridyl-modified gold electrode surface prior to electron transfer. Thus it appears that 4,4'-bipyridyl acts by forming a suitable surface at the electrode-solution interface to which the cytochrome can bind. The binding of the protein to the electrode may therefore be similar in kind to that observed in the cytochrome *c*-oxidase reaction, involving the ϵ -amino groups of lysine residues in the vicinity of the cytochrome *c* heme crevice. The similarities to the cytochrome *c*-oxidase reaction may result from a preferred orientation of the cytochrome when bound at the electrode surface or to cytochrome oxidase such that the exposed heme edge is adjacent to the electrode surface or the cytochrome oxidase, thereby enabling rapid electron transfer to occur.

Hydrated melanin exhibits a reversible switching behavior⁽⁴¹⁾; upon application of an electric field above a certain threshold value this substance changes reversibly and reproducibly from a state of high resistivity to one of low resistivity. The value of the critical switching field depends on the source, nature, and mode of preparation of the pigment; only hydrated specimens show the effect. Cope proposes⁽⁴²⁾ that the high-resistivity form may be considered as an emulsion of electrons in an electron-poor matrix; copious electron injection by virtue of the switching field then inverts the emulsion yielding electron-poor islands in an electron-rich matrix. This should affect the redox potential, and this has indeed been observed.⁽⁴³⁾ The model is further supported by the negative photoconductivity reported⁽⁴⁴⁾ for melanin: this effect, viz., a drop in conductivity upon illumination, is rare in organic materials; it was first reported⁽⁴⁵⁾ in 1965 for chlorpromazine. It is thought to be associated with the presence of centers capable of existing in two different valency, or electronic, states.

5. Charge Transfer and Adsorption. Surface Effects

5.1. Introduction

Surface effects in aqueous media always involve electric double layers characterized by more or less—depending on location—immobilized charges and by a firmly held layer of structured water exhibiting a permittivity considerably below that of its bulk value. There is an extensive literature on this subject⁽⁴⁶⁾; see also the discussion in Pethig's article in this volume.

An adsorbate behaves as an electron donor if the energy of its lowest unoccupied level evaluated in its adsorbed state is situated at or above the Fermi level; the adsorbate, conversely, acts as an acceptor if the highest occupied level of the system substrate plus adsorbate is at or below the Fermi level. Surface complexes are often dimerlike with their energy states well localized within the energy gap, and resulting in a depletion of states in the vicinity of the Fermi levels; any excess charge tends to accumulate on the surface complex,⁽⁴⁷⁾ raising the carrier hopping probability. Some surface complexes consist of an adatom plus its nearest neighbor on the surface, again a dimerlike structure.⁽⁴⁹⁾ The first stage of an adsorption process involves a charge transfer, donor-acceptor, interaction between the adsorbent and the adsorbate,⁽⁴⁸⁾ resulting in an increased effective ionization potential, or electron affinity, of the adsorbed donor or acceptor, and a rise in the local energy levels of the substrate. This stage is followed⁽⁴⁸⁾ by electron, or hole, transfer into stabilized vacant energy levels of the adsorbate. Surface excitons and polaritons are likely to arise on surfaces,⁽⁵⁰⁾ their binding energy is reported to be of the order of 0.1 eV.⁽⁵¹⁾

The topic of surface CTC is basic to the theories of catalysis and is usually treated in that context.⁽⁵²⁾

Surface CTC of chlorpromazine (CPZ) has been shown to be involved⁽⁵³⁾ in hypothermia and sedation of mice; *in vitro* surface adducts between chlorpromazine and collagen have been demonstrated, as well as with lignocain and with phenytoin.⁽⁵⁴⁾ The existence of such CPZ surface adducts has been directly confirmed by electroreflectance measurements.⁽⁹⁴⁾

If an acceptor is incorporated into a cationic micelle, then the rate of reactions involving hydrated electrons is increased by a factor of about 60 times that prevailing at outside negatively charged anionic micelles.⁽⁵⁵⁾

Since counterions interact with, and are attracted by the, say, hydrophilic heads of the colloidal particles, micellar assemblages of such particles should be catalytically active. This, in general, has been found to be the case,⁽⁵⁶⁾ and should be a rewarding field for the study of micellar CTC as well as electrocatalysis. Formation of micellar CTC is also reported to result in considerably shifted ion exchange equilibria.⁽⁵⁷⁾

5.2. Membranes

5.2.1. The Transmembrane Conductance

This topic has been extensively studied and many reviews and treatises dealing with it are available.⁽⁵⁹⁾ In this volume, it is discussed in contributions by Pethig (Chapter 7) and by Habib and Bockris (Chapter 3).

5.2.2. Membrane Surfaces

The surface of a biological membrane exhibits electron donating as well as electron accepting centers, though the surface as a whole is not, in general, electrically neutral, exhibiting the well-known membrane potential. We now wish to suggest that interactions with an external molecular electron acceptor or donor produces a conformational change of the complementary region on the membrane surface, perhaps initiated by Poole-Frenkel, thermally assisted field emission of carriers. The resulting transient voltage pulse may then be reversed by an opposing membrane potential which transfers the carrier back and thus restores the initial steric configuration on the membrane surface.

This hypothesis requires a finite, nonzero, conductance in the surface plane of the membrane because the electron, or hole, initially transferred forms a movable excited surface state, i.e., a polariton.⁽⁶⁰⁾ This may well give rise to a soliton, given a supply of energy at the proper locale.

No direct contact between donor and acceptor is envisaged; charge transfer is expected to occur once the donor approaches the acceptor, or vice versa, within tunneling distance, say 20 Å. The donor-acceptor charge transfer thus yields a pattern of surface charges and of sterically oriented proteins. Moreover, the highly nonuniform, inhomogeneous local electrical fields in the vicinity of the charges produce large, local, mechanical forces which are liable to lead to the opening or closing of ionically conducting transmembrane channels. One could then still further hypothesize that it is this surface charge pattern which is involved in the recognition of "self" and the disturbance of which plays a part in uncontrolled growth, viz., carcinogenesis. It may be significant in this context that of 289 carcinogens listed in the latest, Tenth, Collective Index of Chemical Abstracts, all but one—benzene—are electron acceptors or have pronounced electron accepting sites in their molecular structure.

The transmembrane, transcellular transient mentioned has indeed been observed,⁽⁶¹⁾ e.g., in the hyphae of the water mould *Achyla bisexualis* as well as in other fungi.

The electron involved in the donor-acceptor interaction need not originate from the actual surface layer but may well come from the interior of the bilipid membrane, for example, from intramembrane proteins. The electron escape depth in organic materials is comparable to intermolecular distances.⁽⁶²⁾

Alternatively, it may arise from an "activating ion" which enters into a surface complex with an adsorbed enzyme; it has been shown⁽⁹⁷⁾ that the stability of such complexes is greatly affected by the electric double layer structure of the membrane surface. As the electrostatic screening is

increased, the complex tends to become less stable and the enzyme activity drops.⁽⁹⁷⁾

It is notable that all condensed polycyclic aromatics with a planar molecular structure are carcinogenic—and all have the same and relatively high value of polarization energy, viz., 1.7 eV and thus are capable of acting as efficient electron donors as well as acceptors.⁽⁶³⁾ Whether a given protein is capable of insertion into the membrane is chiefly determined by hydrophobic forces, but the final orientation of the protein depends on electrostatic interactions such as the forces arising from surface charge asymmetries—which are a feature of the hypothesis proposed above—and from local electrochemical events.⁽⁶⁴⁾ Any changes in the electrical surface pattern are likely to propagate to adjacent cell membranes by a process akin to epitaxy, as has been demonstrated for inorganic layers by Distler⁽⁶⁵⁾ for distances of up to about 230 Å. A photoelectret process is said⁽⁶⁵⁾ to be involved.

6. Proton Transfer Complexes

The electron transfer involved in the formation of a charge transfer complex may in suitable systems be coupled to a proton transfer,⁽⁶⁹⁾ resulting in proton transfer complexes. This holds, especially for surface reactions, e.g., at electrodes, and a fortiori in nonaqueous media, because the proton affinity of water is so very high, viz., about 8.9 eV.⁽⁷⁰⁾

The proton affinity A_p is defined as the enthalpy of the reaction



In the gas phase, where there are energy contributions from solvation or polarization, A_p may be measured by means of ion cyclotron spectroscopy.⁽⁸⁰⁾ Proton affinities are often given in kJ/mole; 1 kJ/mole equals 0.0104 eV; 1 kcal/mole equals 0.0436 eV. Exchange repulsion energies do not enter into proton affinities.⁽⁷²⁾ The “relative protonicity” of solvents, akin to the donicity concept for electrons, is discussed by Bayless *et al.*⁽⁷⁴⁾

Hydroxy-dinitro pyridines, e.g., act as electron acceptors and/or as proton donors to, e.g., naphthalene derivatives.⁽⁷¹⁾

The field of protonic CTC has been opened up by Matsunaga and his school,⁽⁶⁹⁾ and has been reviewed by Morokuma.⁽⁷²⁾ This author points to the close linkage existing in such adducts between proton and concurrent electron transfer; one does not occur without the other. The proton complex is said⁽⁷²⁾ to involve an electron transfer from, e.g., an amine to the

proton. There is evidence that the electron transfer in some cases may be catalyzed by the presence of a proton and thus enhanced above its thermally controlled rate.⁽⁷⁵⁾

Proton transfer may, sometimes, be really nothing more than a case of conventional hydrogen bonding, but in many cases the simultaneous transfer of an electron and a proton produces a new different type of adduct; then the difference between mere hydrogen bonding; and complex formation may indeed be dramatic, as pointed out by Arnett and Mitchell.⁽⁷⁶⁾ There is no correlation between the heats of protonation and of hydrogen bonding.⁽⁷⁶⁾

Ion pair and hydrogen bonded complexes⁽⁷⁷⁾ are beyond the scope of this paper. Here, only a few representative examples of protonic complexes can be discussed.

The 1:1 and 1:2 complexes between oxalic acid and ϵ amino acids also involve a proton transfer from the carboxyl group of the oxalic acid to the carboxyl ion of the amino acid; there is a similar interaction in the 1:1 complex between malonic acid and glycine.⁽⁷⁸⁾ These adducts may well involve a great deal of proton delocalization resulting in the formation of proton energy bands rather than energy levels. Proton tunneling is a well-established fact,⁽⁷⁹⁾ and even halogen ions have been shown to be able to tunnel between two suitable energy levels. Ionic charge transfer complexes between benzyl radicals and halide ions have also been reported.⁽⁸⁰⁾

The average time of residence of the proton at a temporary equilibrium site is of the order of one vibrational period of the OH group and of the average lifetime of H_3O^+ , viz., about 2.5×10^{-13} s; since the dielectric relaxation time of water is about 10^{-11} s, it appears that dielectric relaxation and frictional processes are not involved in proton transfer.⁽⁸¹⁾ However, in what appears to be protonic charge transfer complexes between, e.g., aliphatic amines and phenols, a considerable increase in viscosity is reported.⁽⁸²⁾

Intramolecular proton transfer in electronically excited molecules has been reviewed⁽⁸³⁾; in salicylic acid esters it is said to lead to deexcitation of the excited electron⁽⁸⁹⁾; the free energy change involved in that proton transfer is reported as about 0.13–0.22 eV.⁽⁸⁴⁾

The effect has been shown to accompany at least some cases of electronic excitation⁽⁸⁵⁾ in organic molecules, such as aromatic compounds. These effects have hardly been touched upon as far as they affect biologically important transitions, as they are bound to do. This is an important and rather new field which should be explored.

Electron excitation causes only a very small change in the enthalpy of hydrogen bonding; a value of about 0.0044 eV is reported.⁽⁸⁶⁾

Bacteriorhodopsin is a pigment found as a single protein component of the purple membrane of *Halobacterium halobium* and similar extreme

halophiles.⁽⁸⁷⁾ The purple membrane converts light energy by translocating protons across the membrane to generate an electrochemical potential. A review⁽⁸⁸⁾ of this subject has been published. The naturally recurring crystalline structure has been determined with electron microscopy at 7 Å resolution. The protein field is folded into seven α -helical chains, all of which span the hydrophobic core of the membrane.⁽⁸⁸⁾ The chromophore has been located in a lysine residue in a sequence of the second helical chain from the amino terminal group⁽⁸⁹⁾; cf. discussions of methyl-glyoxal-protein complexes. The chromophore's double bond chain makes an angle of about 20° to the plane of the membrane,⁽⁹⁰⁾ with its ionone ring close to the center of the membrane.⁽⁹¹⁾

The role of bacterio-rhodopsin in the proton pumping activity of halobacteria has been further investigated by Bagyinka *et al.*,⁽⁹²⁾ while the light transduction via the pigmented bilipid membranes, of the purple membrane of *H. halobium*, has been studied by Ti Tien.⁽⁹³⁾

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